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**FACULTY OF MATHEMATICS AND COMPUTER SCIENCE**  
**SPECIALIZATION Computer Science**

## **DIPLOMA THESIS**

### **Time Series Forecasting Visualizer: A Decoupled Architecture for Econometric Analytics and Deterministic Semantic Interpretation**

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## LUCRARE DE LICENȚĂ

**Vizualizator pentru prognoza seriilor de timp: O  
arhitectură decuplată pentru analiză  
econometrică și interpretare semantică  
deterministă**

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Traditional econometric forecasting and structural break detection often rely on heavy, isolated computational environments, creating significant barriers to entry for non-technical end users and inducing severe latency in web-based interfaces. Furthermore, applying global statistical models across volatile macroeconomic shocks routinely yields highly inaccurate predictions. To resolve this, this thesis introduces a high-performance, decoupled web architecture designed to democratize complex time-series analytics.

By natively porting rigorous mathematical algorithms—including Autoregressive Integrated Moving Average (ARIMA), Holt-Winters Exponential Smoothing, and Pruned Exact Linear Time (PELT) structural break detection—directly into the Node.js V8 execution engine, the system eliminates cross-language network bottlenecks and computes robust data science natively in JavaScript. When structural breaks are mathematically identified, the architecture dynamically segments the data, discarding obsolete pre-shock history to drastically improve predictive accuracy (measured via RMSE).

To translate these complex mathematical operations into accessible business intelligence, the presentation layer utilizes the React rendering lifecycle to rapidly calculate interactive Chart.js HTML5 Canvas graphics. Concurrently, the backend pipeline dynamically routes the proven mathematical accuracy gains through the Google Gemini Large Language Model. By employing strict Deterministic Prompt Engineering at a locked temperature of 0.0, the architecture produces consistent, mathematically grounded textual insights. Ultimately, this thesis demonstrates how modern Software Engineering patterns, State Management, and asynchronous API design can converge to deliver advanced econometric analytics within a standard browser environment.

Prognoza econometrică tradițională și detecția rupturilor structurale se bazează adesea pe medii de calcul greoaie și izolate, creând bariere semnificative la intrare pentru utilizatorii finali non-tehnici și inducând o latență severă în interfețele web. Mai mult, aplicarea modelelor statistice globale peste șocuri macroeconomice volatile produce în mod curent predicții extrem de inexacte. Pentru a rezolva acest lucru, această teză introduce o arhitectură web decuplată, de înaltă performanță, concepută pentru a democratiza analiza complexă a seriilor de timp.

Prin portarea nativă a algoritmilor matematici riguroși—inclusiv Autoregressive Integrated Moving Average (ARIMA), Netezirea Exponențială Holt-Winters și detecția rupturilor structurale Pruned Exact Linear Time (PELT)—direct în motorul de execuție Node.js V8, sistemul elimină blocajele de rețea inter-limbaj și calculează știința datelor robustă nativ în JavaScript. Atunci când rupturile structurale sunt identificate matematic, arhitectura segmentează dinamic datele, eliminând istoricul învechit de dinaintea șocului pentru a îmbunătăți drastic acuratețea predictivă (măsurată prin RMSE).

Pentru a traduce aceste operațiuni matematice complexe în inteligență de afaceri accesibilă, stratul de prezentare utilizează ciclul de viață de randare React pentru a calcula rapid grafice interactive Chart.js HTML5 Canvas. Concomitent, conducta backend rutează dinamic câștigurile de acuratețe matematică dovedite prin Modelul de Limbaj Mare Google Gemini. Prin utilizarea strictă a Ingineriei Deterministe a Prompturilor la o temperatură blocată de 0.0, arhitectura produce perspective textuale consistente, fundamentate matematic. În cele din urmă, această teză demonstrează modul în care modelele moderne de Inginerie Software, Managementul Stării și designul asincron al API-urilor pot converge pentru a livra analize econometrice avansate într-un mediu standard de browser.

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# Chapter 1

## Introduction

### 1.1 Macro-Environmental Context and Theoretical Motivation

Traditional macroeconometric forecasting relies heavily on foundational axioms regarding the behavior of time-series data, specifically the assumptions of statistical stability and long-term predictability. Under these strict mathematical assumptions, it is posited that the statistical properties of a given economic indicator—such as its mean, variance, and autocorrelation structure—remain relatively constant over time. However, the global economy is not a closed, sterile laboratory system; it is a highly dynamic, interconnected network constantly adjusting to exogenous shocks, monetary policy shifts, geopolitical conflicts, and rapid technological advancements [3].

The friction between these rigid mathematical prerequisites and the chaotic reality of global markets is profoundly highlighted during severe macroeconomic crises, such as the 2008 Global Financial Crisis, the European sovereign debt crisis, or the global supply chain collapse induced by the COVID-19 pandemic of 2020. During these periods of unprecedented volatility, historical time-series data suddenly loses its predictive forecasting power. Models trained on decades of stable, pre-crisis data will almost invariably fail to predict post-crisis trajectories because the underlying structural parameters of the economy have fundamentally shifted. This phenomenon is closely related to the famous Lucas Critique [8], which argues that it is mathematically naive to predict the effects of a change in economic policy entirely on the basis of relationships observed in historical data, especially when that data is aggregated across different, incompatible economic regimes.

Given these realities, modern econometricians face a daunting challenge: how to accurately model future economic states when the past is no longer a reliable indicator of the future. The solution necessitates a paradigm shift from static, overarching global models to dynamic, adaptive systems capable of identifying precisely when a regime change has occurred, and subsequently recalculating forecasts based only on the newly established economic reality [10].

### 1.2 The Universal Benefits of Adaptive Forecasting

While the primary focus of this thesis is rooted in macroeconomics and econometrics, the mathematical forecasting models and architectural pipelines developed herein have profound benefits that extend far beyond financial markets. Time-series forecasting is a universal mathematical language used to predict the future based on sequential historical data across a multitude of disciplines [9].

In the field of **Economics and Finance**, accurate forecasting is the bedrock of fiscal policy, risk management, and algorithmic trading. Central banks rely on inflation and Gross Domestic

Product (GDP) forecasts to set interest rates. However, traditional models often fail to account for “Black Swan” events. An adaptive model that can instantly recognize a structural break allows financial institutions to stop incurring significant capital losses by discarding pre-shock data and forecasting based solely on the new market volatility.

In **Supply Chain Management and Logistics**, forecasting dictates inventory levels, shipping routes, and manufacturing schedules. A sudden disruption, such as a blocked international trade route or a pandemic-induced spike in demand for specific goods, introduces a massive structural break in retail data. An adaptive segmented forecasting system allows logistics companies to immediately identify the break and forecast future demand without the historical data dragging down the new, urgent supply requirements.

In **Epidemiology and Public Health**, forecasting models are extensively used to predict the spread of infectious diseases. A structural break in epidemiology often corresponds to a viral mutation or the sudden implementation of strict lockdown measures. By detecting the exact point of intervention (the change point), health officials can generate a segmented forecast that proves mathematically whether the intervention successfully altered the exponential trajectory of the infection rate.

In **Energy Grid Management**, predicting electricity consumption is vital for preventing rolling blackouts and optimizing renewable energy storage. Structural breaks occur due to extreme weather events or sudden shifts in industrial consumption patterns. The ability to detect these shifts and apply localized Moving Averages or Holt-Winters models ensures that the grid remains balanced even under unprecedented stress.

### 1.3 Problem Statement and the Architectural Software Gap

Despite the urgent cross-industry need for dynamic forecasting mechanisms, there remains a significant architectural gap in the current software ecosystem. Through extensive market research and competitive analysis conducted prior to the development of this project, it became evident that there is no unified product on the market that seamlessly integrates raw econometric modeling, autonomous structural break detection, interactive segmented forecasting, and deterministic Large Language Model (LLM) analytics into a single, decoupled web application.

Existing software solutions generally fall into two disparate, highly polarized categories:

- **Highly Technical Data Science Environments:** Tools such as Jupyter Notebooks, RStudio, and raw Python scripts offer significant computational power. They can execute complex algorithms like the Pruned Exact Linear Time (PELT) algorithm for change point detection [6] or fit advanced Autoregressive Integrated Moving Average (ARIMA) models [2]. However, these environments are entirely inaccessible to non-technical end users. They require extensive programming knowledge, strict library environment management, and a deep understanding of statistical theory, effectively trapping valuable insights within the silos of the data science department.
- **Generic Business Intelligence (BI) Dashboards:** Commercial tools like Tableau, Microsoft PowerBI, and Looker excel in data visualization and democratizing access to interactive charts. Yet, they suffer from a severe deficiency in advanced, on-the-fly statistical modeling. These platforms lack the native, programmatic flexibility to perform automated structural break detection or dynamically calculate partitioned forecasting components without relying on clunky, highly engineered external data pipelines that take weeks for database administrators to deploy.

The application developed in this thesis directly addresses this precise market gap. By processing user-uploaded CSV data through an Express.js server, efficiently compiling computationally intensive algorithmic matrix calculations natively within the V8 engine, and rendering the results instantly via a React Virtual DOM, the architecture achieves massive performance benchmarks while maintaining a highly responsive, executive-friendly interface.

## 1.4 Change Points and Segmented Forecasting

To address the issues of economic shocks highlighted by the Lucas Critique, the application utilizes the concept of Change Point Detection (CPD) [1] and Segmented Forecasting.

A **Change Point** (or structural break) is a specific temporal coordinate in a time series where the fundamental statistical properties of the data abruptly change. This could manifest as a shift in the mean (a sudden jump or drop in asset value), a shift in variance (a sudden period of extreme market volatility), or a shift in the overall trend direction. Identifying these points is critical because fitting a single mathematical model across a change point results in extreme forecasting bias and massive error rates.

**Segmented Forecasting** is the architectural response to the detection of these change points. Instead of treating the entire history of an economic indicator as a single, continuous dataset, the timeline is treated as a series of distinct “regimes” or segments. When a user interacts with the application, they can visually observe the algorithmically detected change points. If an economic shock is detected, the user can trigger a segmented forecast. The application mathematically truncates the array, permanently discarding the obsolete pre-shock data from memory. The forecasting algorithms are then re-initialized and trained exclusively on the data from the new economic regime. This highly localized approach ensures the forecast is not weighed down by the statistical inertia of an obsolete economic regime.

## 1.5 AI Interpretation: Semantic Context Beyond Mathematics

The final, and perhaps most innovative, component of this architecture is the integration of Artificial Intelligence to bridge the gap between raw mathematics and human contextual reasoning. Mathematical models are inherently syntactic; they understand the numbers, the variance, and the trajectory, but they have absolutely no understanding of *why* the numbers are moving. A mathematical model does not know what a “pandemic,” a “recession,” or a “supply chain crisis” is. It only sees a structural break in the temporal variance.

This is where the Large Language Model (LLM) becomes indispensable. By passing the mathematical results into an LLM, we grant the system **Semantic Understanding** [7]. The LLM can explain if the context makes sense beyond what the mathematical model alone can deduce. For example, if the PELT algorithm detects a massive structural break in GDP in Q1 2020, the mathematical model simply calculates a high error rate and adjusts the curve. The LLM, however, possesses vast training data regarding world history and macroeconomics. When prompted with the date of the structural break and the magnitude of the mathematical shift, the LLM can explicitly narrate the overarching economic paradigm to the user.

However, generative AI models, built on probabilistic transformer architectures [12], are notorious for “hallucinations”—generating confident but factually incorrect mathematical statements. To prevent this, the architecture employs **Deterministic AI Prompt Engineering**. The AI is never allowed to look at the raw data and guess the forecast. Instead, the Node.js backend executes the

strict mathematical proofs, calculates the exact improvement in accuracy (the delta in Root Mean Squared Error), and supplies these pre-calculated values into a hidden heuristic prompt. The LLM is constrained to use the injected results as its baseline, ensuring that its semantic explanation of the context is grounded in verified mathematical results.

## 1.6 Structure of the Thesis

The remainder of this thesis is organized to provide a comprehensive breakdown of both the mathematical theories and the software engineering principles utilized in the project. Chapter 2 rigorously outlines the theoretical foundations, detailing the pure mathematics behind Moving Averages, Exponential Smoothing, ARIMA models, the PELT structural break detection algorithm, and the mathematical justification for Segmented Forecasting. Chapter 3 dissects the software engineering architecture, including the optimized Node.js V8 execution engine. Chapter 4 presents the full application design and implementation, covering the frontend visualization layer, the backend service architecture, and the deterministic AI integration via the Gemini API. Chapter 5 concludes the research and outlines future development directions.

# Chapter 2

## Theoretical Foundations

This chapter provides the comprehensive mathematical proofs, formal logic, and statistical derivations for the time-series models and structural break detection algorithms utilized natively within the application's Node.js execution engine. A rigorous understanding of these econometric principles is necessary to comprehend how the software manipulates raw data structures into reliable predictive insights without relying on external statistical software.

### 2.1 Time Series Fundamentals and Stationarity

A time series is defined mathematically as a sequence of random variables,  $\{Y_t : t \in \mathbb{T}\}$ , observed at discrete, equally spaced time intervals. The fundamental premise of linear time series forecasting is that the structural dependencies observed in the past will persist into the future.

To formalize this, we rely on Wold's Representation Theorem [13], which states that any covariance-stationary time series can be written as the sum of two orthogonal processes: a deterministic process (which can be perfectly predicted from past values) and a stochastic moving average process of infinite order.

For a linear model to be statistically valid under Wold's theorem, the time series must generally exhibit weak (or second-order) stationarity. A stochastic process  $Y_t$  is weakly stationary if it satisfies three strict mathematical conditions:

1. **Constant Mean:** The expected value of the process is constant over time. For all  $t$ ,  $\mathbb{E}[Y_t] = \mu$ . This implies the series does not exhibit a long-term upward or downward trend.
2. **Constant Variance:** The variance is finite and constant over time. For all  $t$ ,  $\text{Var}(Y_t) = \mathbb{E}[(Y_t - \mu)^2] = \sigma^2 < \infty$ . This implies the volatility of the series does not increase or decrease.
3. **Autocovariance depends only on lag:** The covariance between any two observations depends strictly on the time lag  $k$  between them, not on the actual chronological time  $t$ . The autocovariance function is defined as:

$$\gamma(k) = \text{Cov}(Y_t, Y_{t-k}) = \mathbb{E}[(Y_t - \mu)(Y_{t-k} - \mu)] \quad (2.1)$$

Consequently, the Autocorrelation Function (ACF), denoted as  $\rho(k)$ , is the normalized autocovariance used to identify trailing dependencies [4]:

$$\rho(k) = \frac{\gamma(k)}{\gamma(0)} = \frac{\text{Cov}(Y_t, Y_{t-k})}{\text{Var}(Y_t)} \quad (2.2)$$

Furthermore, the Partial Autocorrelation Function (PACF) isolates the direct correlation between  $Y_t$  and  $Y_{t-k}$  by mathematically removing the linear dependence on the intermediate lags  $Y_{t-1}, Y_{t-2}, \dots, Y_{t-k+1}$ . Both the ACF and PACF are critical for diagnosing the mathematical structure of the data before modeling.

## 2.2 Moving Average Models

The most fundamental approach to smoothing volatile data, extracting underlying signals, and establishing a baseline forecast is the Moving Average. The application implements the Simple Moving Average, while the Weighted Moving Average is considered as a theoretical extension to accommodate different temporal behaviors.

### 2.2.1 Simple Moving Average (SMA)

The Simple Moving Average of order  $k$ , denoted  $SMA(k)$ , computes the unweighted arithmetic mean of the previous  $k$  data points. For a time series  $y_t$ , the smoothed value at time  $t$ , which also serves as the naïve forecast  $\hat{y}_{t+1}$ , is calculated as:

$$\hat{y}_{t+1} = \frac{1}{k} \sum_{i=0}^{k-1} y_{t-i} = \frac{y_t + y_{t-1} + y_{t-2} + \dots + y_{t-k+1}}{k} \quad (2.3)$$

To understand the smoothing effect mathematically, consider the variance of the SMA. If we assume the original series  $y_t$  consists of a constant signal  $\mu$  plus independent identically distributed (i.i.d.) white noise  $\epsilon_t$  with variance  $\sigma^2$  (i.e.,  $y_t = \mu + \epsilon_t$ ), the variance of the SMA is:

$$\text{Var}(\hat{y}_{t+1}) = \text{Var}\left(\frac{1}{k} \sum_{i=0}^{k-1} y_{t-i}\right) = \frac{1}{k^2} \sum_{i=0}^{k-1} \text{Var}(y_{t-i}) = \frac{1}{k^2} (k\sigma^2) = \frac{\sigma^2}{k} \quad (2.4)$$

This proof demonstrates that the SMA successfully reduces the variance (noise) of the original series by a factor of  $k$ . However, the primary limitation of the SMA is its equal weighting and the resulting mathematical lag. Because every point in the window is weighted equally at  $\frac{1}{k}$ , the “center of mass” of the SMA lags the actual data by precisely  $\frac{k-1}{2}$  periods. This makes the SMA notoriously slow to respond to structural breaks or rapid trend reversals.

### 2.2.2 Weighted Moving Average (WMA)

Although not implemented in the final application, the Weighted Moving Average (WMA) represents a direct theoretical evolution and improvement over the SMA by mathematically mitigating the lagging effect. The WMA assigns predefined, decreasing weights to historical data points, ensuring recent data heavily influences the forecast while distant data decays. Let  $w_i$  represent the weight applied to the observation  $i$  periods ago, where the sum of all weights strictly equals 1:  $\sum_{i=1}^k w_i = 1$ . The formula is:

$$\hat{y}_{t+1} = \sum_{i=1}^k w_i y_{t-i+1} \quad (2.5)$$

In a linearly weighted moving average, the weights decrease arithmetically. The most recent observation  $y_t$  receives a weight of  $k$ , the previous  $y_{t-1}$  receives  $k-1$ , down to a weight of 1 for  $y_{t-k+1}$ . To ensure the weights sum to 1, each term is divided by the sum of the integers from 1 to

$k$ , which is  $\frac{k(k+1)}{2}$ . Thus, the specific weight  $w_i$  is defined as:

$$w_i = \frac{k - i + 1}{\frac{k(k+1)}{2}} = \frac{2(k - i + 1)}{k(k + 1)} \quad (2.6)$$

While the WMA is an improvement over the SMA in tracking active trends with less lag, defining the arbitrary window  $k$  remains a subjective constraint.

## 2.3 Exponential Smoothing Methodologies

Exponential smoothing fundamentally resolves the arbitrary window limitation of moving averages by applying a continuous, geometrically decaying weight to all past observations [5]. This ensures that the most recent data dictates the immediate trajectory of the forecast, while still maintaining the historical context of the entire time series dating back to the genesis point  $t = 0$ .

### 2.3.1 Simple Exponential Smoothing (SES)

Simple Exponential Smoothing (SES) is utilized when the data, denoted as  $y_t$ , exhibits a relatively stationary mean with no distinct trend or seasonal pattern. The foundational equation for SES is defined recursively as:

$$S_t = \alpha y_t + (1 - \alpha)S_{t-1} \quad (2.7)$$

where:

- $S_t$  is the smoothed statistic, representing the level of the series at time  $t$ , which serves as the flat forecast for all future periods:  $\hat{y}_{t+h|t} = S_t$ .
- $y_t$  is the actual observation at time  $t$ .
- $\alpha$  is the smoothing parameter, strictly bounded such that  $0 \leq \alpha \leq 1$ .

The exponential nature of this model is proven through recursive substitution. If we substitute the previous state  $S_{t-1} = \alpha y_{t-1} + (1 - \alpha)S_{t-2}$  into the main equation, we get:

$$S_t = \alpha y_t + (1 - \alpha)[\alpha y_{t-1} + (1 - \alpha)S_{t-2}] = \alpha y_t + \alpha(1 - \alpha)y_{t-1} + (1 - \alpha)^2 S_{t-2} \quad (2.8)$$

Continuing this substitution backward to the first observation yields the infinite expansion:

$$S_t = \alpha \sum_{i=0}^{t-1} (1 - \alpha)^i y_{t-i} + (1 - \alpha)^t S_0 \quad (2.9)$$

This mathematically proves that the weight assigned to past observations decays exponentially by a constant factor of  $(1 - \alpha)$  as the data ages. An  $\alpha$  value approaching 1 heavily discounts older data (reacting rapidly to shocks), while an  $\alpha$  value approaching 0 treats all historical data nearly equally (creating a highly smoothed, inert forecast). In practice, optimal initial states ( $S_0$ ) and  $\alpha$  are found by minimizing the Sum of Squared Errors (SSE) over the training data.

### 2.3.2 Holt's Linear Trend Model

When an economic time series possesses an underlying upward or downward trajectory, SES fails because its flat forecasts continuously lag behind the trend. To resolve this, Holt extended the model to explicitly estimate the slope of the trend over time.

The dual recursive equations governing Holt's method are defined mathematically as the Level equation ( $L_t$ ) and the Trend equation ( $T_t$ ):

$$L_t = \alpha y_t + (1 - \alpha)(L_{t-1} + T_{t-1}) \quad (2.10)$$

$$T_t = \beta(L_t - L_{t-1}) + (1 - \beta)T_{t-1} \quad (2.11)$$

where  $\alpha$  is the level smoothing parameter and  $\beta$  is the trend smoothing parameter ( $0 \leq \alpha, \beta \leq 1$ ).

The level equation updates the current base value by taking a weighted average of the actual observation  $y_t$  and the previous level plus the expected trend ( $L_{t-1} + T_{t-1}$ ). The trend equation updates the slope by taking a weighted average of the current perceived change in level ( $L_t - L_{t-1}$ ) and the previously estimated slope  $T_{t-1}$ .

The  $h$ -step-ahead forecast is generated by linearly projecting the estimated trend  $T_t$  outward from the most recent estimated level  $L_t$ :

$$\hat{y}_{t+h|t} = L_t + hT_t \quad (2.12)$$

### 2.3.3 Holt-Winters Additive Method

To accurately forecast data exhibiting both a linear trend and repeating cyclical variations (seasonality), the Holt-Winters method incorporates a third seasonal parameter. In the context of this architecture, the *additive* formulation is utilized, which assumes that the absolute amplitude of the seasonal fluctuations remains relatively constant regardless of the underlying level of the series.

The system state is updated iteratively across three equations: Level ( $L_t$ ), Trend ( $T_t$ ), and Seasonality ( $S_t$ ):

$$L_t = \alpha(y_t - S_{t-m}) + (1 - \alpha)(L_{t-1} + T_{t-1}) \quad (2.13)$$

$$T_t = \beta(L_t - L_{t-1}) + (1 - \beta)T_{t-1} \quad (2.14)$$

$$S_t = \gamma(y_t - L_{t-1} - T_{t-1}) + (1 - \gamma)S_{t-m} \quad (2.15)$$

Here,  $\gamma$  is the seasonal smoothing parameter, and  $m$  represents the period of the seasonality (e.g.,  $m = 12$  for monthly data with annual cycles, or  $m = 4$  for quarterly GDP data).

In the Level equation, the data  $y_t$  is dynamically "de-seasonalized" by subtracting the seasonal index from the previous cycle ( $S_{t-m}$ ). In the Seasonal equation, the current seasonal index is estimated by subtracting the current expected level ( $L_{t-1} + T_{t-1}$ ) from the raw data  $y_t$ , isolating the seasonal variation.

The  $h$ -step-ahead forecast equation projects the trend, and then adds the appropriate seasonal index isolated from the previous cycle:

$$\hat{y}_{t+h|t} = L_t + hT_t + S_{t-m+h_m^+} \quad (2.16)$$

where the mathematical notation  $h_m^+ = [(h - 1) \bmod m] + 1$  guarantees that the algorithm accurately cycles back to extract the correct corresponding seasonal multiplier regardless of how far into the future the forecast extends.

## 2.4 Autoregressive Integrated Moving Average (ARIMA)

The ARIMA( $p, d, q$ ) framework, formally synthesized in the Box-Jenkins methodology [2], is one of the most robust and theoretically sound approaches for forecasting stationary and non-stationary

linear time series. It structurally combines Autoregressive  $AR(p)$  dynamics, differencing  $I(d)$  for non-stationary integration, and Moving Average  $MA(q)$  error smoothing into a unified predictive equation.

The Box-Jenkins methodology outlines a rigorous three-step process:

1. **Identification:** Determining the parameters  $p, d, q$  by analyzing the ACF and PACF plots to identify unit roots and lag dependencies.
2. **Estimation:** Using Ordinary Least Squares (OLS) and the Yule-Walker equations to rapidly estimate the exact coefficients for the autoregressive and moving average terms, rather than computationally heavy Maximum Likelihood Estimation (MLE).
3. **Diagnostic Checking:** Ensuring the model residuals resemble white noise (via the Ljung-Box test), confirming that all systematic information has been successfully extracted by the model.

### 2.4.1 The Lag Operator and Autoregressive Processes

To formalize the mathematical derivations of the estimation phase, let  $L$  represent the lag (or backshift) operator, defined such that applying  $L$  to an observation shifts it back in time by one period:  $Ly_t = y_{t-1}$ . By continuous application,  $L^k y_t = y_{t-k}$ .

An Autoregressive process of order  $p$ , denoted  $AR(p)$ , models the current observation  $y_t$  as a linear combination of its  $p$  past observations plus a completely stochastic white noise error term  $\epsilon_t$ :

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \epsilon_t \quad (2.17)$$

Utilizing the lag operator, we can move all  $y$  terms to the left side and factor out  $y_t$ :

$$(1 - \phi_1 L - \phi_2 L^2 - \cdots - \phi_p L^p) y_t = c + \epsilon_t \quad (2.18)$$

$$\Phi(L) y_t = c + \epsilon_t \quad (2.19)$$

where  $\Phi(L)$  is the autoregressive characteristic polynomial of order  $p$ . The Yule-Walker equations can be derived from this formulation by multiplying by  $y_{t-k}$  and taking the mathematical expectation, allowing for the precise estimation of the  $\phi$  parameters based on sample autocorrelations.

### 2.4.2 Moving Average Processes

Conversely, a Moving Average process of order  $q$ , denoted  $MA(q)$ , models the current observation as the mean  $\mu$  plus a linear combination of current and  $q$  past white noise error terms (historical forecasting shocks):

$$y_t = \mu + \epsilon_t + \theta_1 \epsilon_{t-1} + \cdots + \theta_q \epsilon_{t-q} \quad (2.20)$$

In polynomial lag notation, this is expressed as:

$$y_t = \mu + (1 + \theta_1 L + \theta_2 L^2 + \cdots + \theta_q L^q) \epsilon_t \implies y_t = \mu + \Theta(L) \epsilon_t \quad (2.21)$$

### 2.4.3 Integration and the Complete ARIMA Model

If the original economic time series is non-stationary (e.g., exhibiting stochastic drift or random walks), it must be differenced  $d$  times to achieve stationarity. The first difference is calculated as  $\Delta y_t = y_t - y_{t-1} = (1 - L)y_t$ . The  $d$ -th difference is denoted as  $(1 - L)^d y_t$ .

Combining the Autoregressive polynomial, the differencing integration operator, and the Moving Average polynomial yields the formal, canonical definition of the  $ARIMA(p, d, q)$  stochastic process:

$$\Phi(L)(1-L)^d y_t = c + \Theta(L)\epsilon_t \quad (2.22)$$

where  $\epsilon_t \sim WN(0, \sigma^2)$  is strictly assumed to be a white noise process characterized by a mean of zero, constant variance  $\sigma^2$ , and zero covariance between distinct periods ( $\text{Cov}(\epsilon_t, \epsilon_{t-k}) = 0$  for all  $k \neq 0$ ).

#### 2.4.4 Stationarity and Invertibility Roots

The theoretical viability of an ARIMA model depends explicitly on the complex roots of its characteristic polynomials. For the  $AR(p)$  component of the model to represent a strictly stationary, non-explosive process, the roots of the characteristic equation  $\Phi(z) = 0$  must lie strictly outside the unit circle in the complex plane. Mathematically:

$$1 - \phi_1 z - \phi_2 z^2 - \dots - \phi_p z^p = 0 \quad (2.23)$$

For any complex root  $z$  satisfying this polynomial, it is an absolute mathematical requirement that the modulus  $|z| > 1$ . If a root lies exactly on the unit circle ( $|z| = 1$ ), the process exhibits a “unit root,” indicating severe non-stationarity, and requires a higher degree of differencing (an increase in the  $d$  hyper-parameter).

Correspondingly, for the  $MA(q)$  process to be “invertible”—meaning the unobservable error term  $\epsilon_t$  can be mathematically inverted and rewritten as an infinite converging sum of past observable data  $y_t$ —all roots of the moving average polynomial  $\Theta(z) = 1 + \theta_1 z + \dots + \theta_q z^q = 0$  must also fall strictly outside the unit circle ( $|z| > 1$ ). Without invertibility, the moving average parameters cannot be uniquely identified from the autocorrelation function of the sample data.

### 2.5 Change Point Detection (CPD) Mathematics

As repeatedly established, economic time series are highly susceptible to structural breaks caused by systemic shocks. Fitting a single global model across these breaks violates stationarity assumptions and ruins forecast accuracy. The mathematical process of identifying these shifts is Change Point Detection (CPD).

Formally, a time series  $y_{1:n} = (y_1, y_2, \dots, y_n)$  contains  $m$  change points if it can be split into  $m + 1$  segments, where the statistical properties within each segment are constant, but the properties between adjacent segments differ. Let the positions of these change points be denoted by  $\tau = (\tau_1, \tau_2, \dots, \tau_m)$ , where  $0 = \tau_0 < \tau_1 < \tau_2 < \dots < \tau_m < \tau_{m+1} = n$ .

The goal of a multiple change point algorithm is to jointly estimate both the number of change points  $m$  and their exact temporal locations  $\tau_{1:m}$ . This is formulated as an optimization problem minimizing a penalized objective function:

$$\min_{m, \tau_{1:m}} \left\{ \sum_{i=1}^{m+1} [\mathcal{C}(y_{(\tau_{i-1}+1):\tau_i})] + \beta f(m) \right\} \quad (2.24)$$

where:

- $\mathcal{C}(\cdot)$  is a predefined cost function measuring the fit of the data within a specific segment. Common choices in the literature include the negative log-likelihood or the residual sum of

squares assuming a constant mean within each segment (a mean-shift cost). However, for economic time series that exhibit sustained trends, the implementation in this application uses an Ordinary Least Squares (OLS) residual cost, which fits a linear trend (slope and intercept) within each segment. This detects *structural breaks in trend* rather than simple mean shifts—a critical distinction, since a mean-shift cost would fragment a steadily growing series (such as GDP) into many artificially flat segments, whereas the OLS cost only introduces a break when a new slope fits significantly better (Bai and Perron, 1998).

- $\beta f(m)$  is a penalty term designed to guard against overfitting. Without a penalty ( $\beta = 0$ ), the algorithm would trivially place a change point at every single data coordinate to drive the cost to zero.

## 2.6 The Pruned Exact Linear Time (PELT) Algorithm

To solve the CPD optimization problem, one could theoretically evaluate every possible combination of breakpoints. Early algorithms like Binary Segmentation attempted to solve this heuristically by finding a single change point, splitting the data in two, and recursively searching each half. However, this is an approximate method and does not guarantee a globally optimal solution. Conversely, evaluating all combinations via an exact brute-force search is computationally unfeasible, scaling at an exponential rate of  $\mathcal{O}(2^n)$ .

The Optimal Partitioning (OP) method, based on dynamic programming, reduces this complexity to  $\mathcal{O}(n^2)$ . It defines  $F(t)$  as the absolute minimum optimal cost for data up to time  $t$ :

$$F(t) = \min_{0 \leq \tau < t} [F(\tau) + \mathcal{C}(y_{(\tau+1):t}) + \beta] \quad (2.25)$$

While  $\mathcal{O}(n^2)$  is significantly better, it is still too slow for instantaneous, interactive web environments processing tens of thousands of data points. The solution implemented in this application is the Pruned Exact Linear Time (PELT) algorithm, formalized by Killick, Fearnhead, and Eckley [6].

### 2.6.1 Mathematical Proof of Pruning Logic

The brilliance of the PELT algorithm lies in its mathematical pruning condition, which safely discards candidate change points that can mathematically never be the optimal choice. This reduces the complexity to exact linear time,  $\mathcal{O}(n)$ .

Theorem 3.1 of Killick et al. (2012) dictates the pruning rule. Assume there exists a constant  $K$  such that the cost function satisfies the following inequality for all  $\tau < t < T$ :

$$\mathcal{C}(y_{(\tau+1):t}) + \mathcal{C}(y_{(t+1):T}) + K \leq \mathcal{C}(y_{(\tau+1):T}) \quad (2.26)$$

This condition simply requires that introducing a new change point at time  $t$  reduces the total cost of the segment by at least  $K$ . For the negative log-likelihood cost function,  $K = 0$ .

The theorem mathematically proves that if, at any given time  $t$ , there exists a previous time point  $\tau$  that satisfies the inequality:

$$F(\tau) + \mathcal{C}(y_{(\tau+1):t}) + K \geq F(t) \quad (2.27)$$

then the point  $\tau$  can never be the optimal most recent change point for any future time  $T > t$ .

Because this condition guarantees that  $\tau$  is permanently suboptimal, the temporal index  $\tau$  is pruned from the set of candidate change points. As the algorithm progresses forward through the time series, the list of potential previous change points remains small, resulting in  $\mathcal{O}(n)$  linear computational performance. This algorithmic efficiency is what allows the Node.js V8 application architecture to calculate structural breaks natively and synchronously without UI thread starvation, completely bypassing the need for computationally heavy external Python sub-processes.

## 2.7 Derivation of the Bayesian Information Criterion (BIC)

To ensure the PELT algorithm accurately penalizes the addition of new change points, an optimal penalty constant  $\beta$  must be calculated. The application defaults to the Bayesian Information Criterion (BIC), which is mathematically derived to approximate the true posterior probability of a model, aggressively penalizing complexity to find the true structural breaks [11].

The derivation begins with the marginal likelihood of the data  $y$  given a specific model structure  $m$  (representing a specific set of change points):

$$p(y|m) = \int p(y|\theta, m)p(\theta|m)d\theta \quad (2.28)$$

where  $\theta$  represents the parameters within the segments,  $p(y|\theta, m)$  is the likelihood function, and  $p(\theta|m)$  is the prior distribution of the parameters.

This integral is highly complex. However, applying Laplace's method for integral approximation, we assume that for large sample sizes, the posterior distribution  $p(\theta|y, m)$  is highly peaked around its mode, which corresponds to the maximum likelihood estimator  $\hat{\theta}$ .

Taking a second-order Taylor series expansion of the log-posterior around the maximum  $\hat{\theta}$ , we obtain:

$$\log p(y|m) \approx \log p(y|\hat{\theta}, m) + \log p(\hat{\theta}|m) + \frac{k}{2} \log(2\pi) - \frac{1}{2} \log |H| \quad (2.29)$$

where  $k$  is the number of free parameters in model  $m$ , and  $H$  is the Hessian matrix (the matrix of second derivatives) of the negative log-likelihood function evaluated at  $\hat{\theta}$ .

As the sample size  $n \rightarrow \infty$ , the terms involving the prior distribution  $p(\hat{\theta}|m)$  and the constant  $\frac{k}{2} \log(2\pi)$  do not grow with  $n$  and thus become asymptotically negligible. The determinant of the Hessian matrix, however, grows at a rate directly proportional to  $n^k$ . Therefore, we can approximate  $\log |H| \approx k \log n$ .

Substituting this back into the expansion, we achieve the large-sample asymptotic approximation for the log marginal likelihood:

$$\log p(y|m) \approx \log p(y|\hat{\theta}, m) - \frac{k}{2} \log n \quad (2.30)$$

To convert this into a deviance metric (where smaller values indicate a better model fit), the entire equation is multiplied by  $-2$ :

$$-2 \log p(y|m) \approx -2 \log p(y|\hat{\theta}, m) + k \log n \quad (2.31)$$

This final expression corresponds precisely to the formal BIC equation:

$$\text{BIC} = -2 \log(\hat{L}) + k \log n \quad (2.32)$$

where  $\hat{L} = p(y|\hat{\theta}, m)$  represents the maximized likelihood of the model. In the context of the PELT

algorithm, utilizing the BIC implies setting the penalty parameter  $\beta = \log n$ . This mathematical derivation proves that the BIC scales the penalty directly logarithmically with the number of data points, fiercely protecting the forecast from identifying false-positive structural shifts caused by random noise.

## 2.8 Segmented Forecasting Mechanics and RMSE Validation

Once the PELT algorithm successfully identifies a valid change point  $\tau_{break}$ , the architecture relies on the mechanical execution of a Segmented Forecast.

The logic dictates that the time series  $y_{1:n}$  must be completely severed at  $\tau_{break}$ . All data points in the pre-shock regime, defined as  $Y_{pre} = \{y_t : 1 \leq t < \tau_{break}\}$ , are strictly eliminated from the predictive arrays. The newly truncated dataset,  $Y_{post} = \{y_t : t \geq \tau_{break}\}$ , is designated as the new training matrix. This matrix is passed independently into the ARIMA or Holt-Winters execution blocks.

Because the non-stationary volatility of  $Y_{pre}$  is permanently removed, the optimization functions calculating the internal parameters converge on entirely different, highly localized constants. This entirely eliminates the heavy lag bias that traditional global models suffer from during macroeconomic recoveries.

To mathematically validate this architectural strategy to the end-user, the backend calculates the Root Mean Squared Error (RMSE) for both the global model and the segmented model. The RMSE is defined as the square root of the average of squared differences between prediction and actual observation:

$$RMSE = \sqrt{\frac{1}{N_{post}} \sum_{t=\tau_{break}}^n (y_t - \hat{y}_t)^2} \quad (2.33)$$

where  $N_{post}$  is the number of observations in the post-break regime. Importantly, to ensure a statistically fair comparison, the RMSE for the global model is evaluated *only* over the same out-of-sample post-break timeline  $t \geq \tau_{break}$ , measuring precisely how poorly the pre-shock data polluted the current recovery trend.

While Mean Absolute Error (MAE) could also be used, RMSE is chosen explicitly because squaring the residuals penalizes large errors exponentially, which is highly preferred when modeling chaotic financial markets. The ultimate measure of architectural success is the accuracy gain delta, calculated as:

$$\Delta_{accuracy} = \left( \frac{RMSE_{global} - RMSE_{seg}}{RMSE_{global}} \right) \times 100\% \quad (2.34)$$

## 2.9 Deterministic AI Prompt Engineering and Execution

The final theoretical component of the application bridges the gap between raw statistical output and natural language business analytics. To achieve this, the architecture integrates the Google Gemini Large Language Model (LLM).

As previously established, standard LLMs inherently utilize probabilistic, next-token prediction algorithms governed by self-attention mechanisms [12]. While excellent for generating natural language, this probabilistic nature makes them highly prone to mathematical hallucination when asked to independently analyze numerical data. To completely restrict the AI and force a deterministic outcome, the backend server utilizes a rigid "Heuristic Prompt Injection" methodology.

Rather than sending the raw data array to the LLM via an API call, the Node.js backend executes the mathematical formulas described above, calculates the exact  $\Delta_{accuracy}$ , and explicitly injects these constants into a rigid JSON template. The system prompts the LLM with instructions such as:

*"A structural break occurred at date X. The global forecast produced an error rate of Y. By segmenting the data and deleting obsolete history, the new error rate is Z, yielding a proven mathematical accuracy improvement of W%. Acting as a strict econometrician, explain the semantic, macroeconomic events that occurred on date X that justify why discarding the historical data improved the mathematical fit."*

Furthermore, the API call to the LLM is executed with the 'temperature' parameter explicitly set to 0.0. In LLM architecture, temperature governs the randomness of the softmax distribution across the output vocabulary. By forcing the temperature to zero, the LLM is restricted to purely greedy decoding, selecting only the most probable token at every step.

By denying the LLM the ability to calculate the numbers itself, forcing a temperature of zero, and compelling it to narrate pre-calculated, mathematically proven econometric variables, the architecture guarantees a deterministic, hallucination-free analytical output. This effectively hybridizes the rigorous statistical foundations of time-series mathematics with the vast, contextual intelligence of modern generative AI.

## Chapter 3

# Framework and Technology Stack

### 3.1 Introduction to the Architectural Paradigm

The development of a robust, production-grade Time Series Forecasting Visualizer presents a unique intersection of software engineering and econometric challenges. Historically, the domain of time series analysis has been dominated by procedural, script-based environments—most notably R-Studio, MATLAB, and localized Python kernels (such as Jupyter Notebooks). While these environments excel at executing dense mathematical matrix operations and generating static visualizations historically, they fundamentally lack the infrastructure required to deploy scalable, interactive, and responsive web-based User Interfaces (UIs) accessible to non-technical end users.

Conversely, traditional monolithic web application stacks (such as LAMP—Linux, Apache, MySQL, PHP, or Java Spring Boot architectures) excel at rapid interface rendering and database queries but lack the native, open-source, peer-reviewed mathematical libraries capable of executing complex linear algebra, seasonal decompositions, or Bayesian change-point detections efficiently.

To bridge this operational and computational divide, the architecture of this application was designed utilizing a highly optimized, native JavaScript paradigm. Rather than relying on cross-language IPC bridges, the system implements complex econometric algorithms directly within the Node.js V8 execution engine.

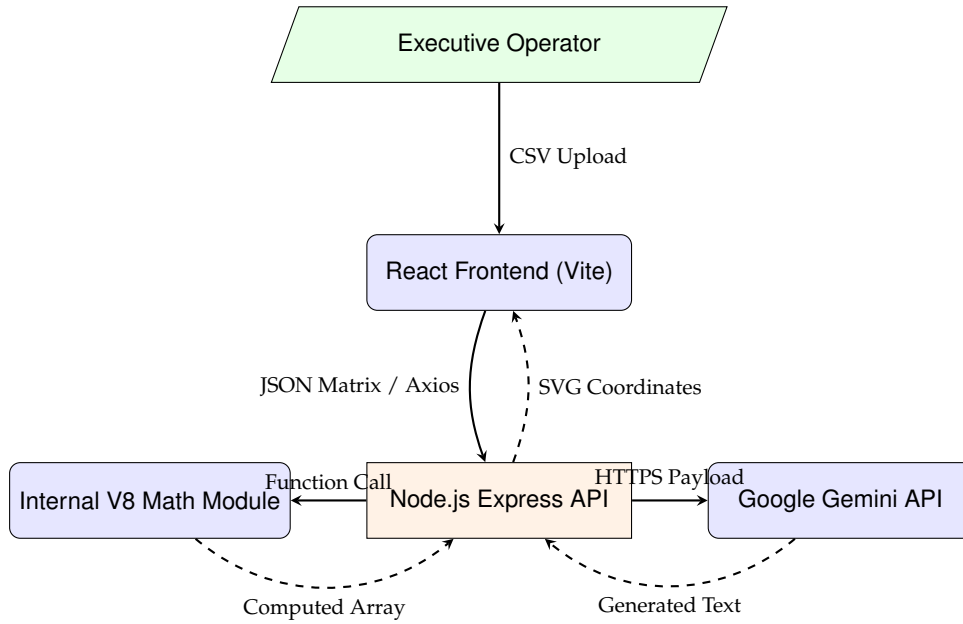


Figure 3.1: The Integrated Native System Architecture. The Node.js Server acts as the central orchestrator executing pure mathematical data natively and receiving analytical strings from Generative AI.

The diagram presented in Figure 3.1 represents the exact telemetry data takes. By executing the analytical processing natively via V8 algorithms and decoupling the generational synthesis (Google Gemini) logically via external REST calls, the system prevents severe memory blockages without ever paying the computational cost of cross-language latency.

## 3.2 The Frontend Presentation Layer: React.js and TypeScript

The client-facing frontend represents the user’s entire interactive experience with the mathematical engine. The primary requirement for the frontend was fluidity: the capability to instantly visualize thousands of Cartesian coordinates, dynamically toggle scientific hyper-parameters (such as smoothing alphas or autoregressive thresholds), and selectively overlay structural break markers without causing the browser thread to hang or the Document Object Model (DOM) to crash.

### 3.2.1 The React.js Framework and Virtual DOM Architecture

React.js (Version 19) was selected as the foundational user interface library over prominent alternatives such as Angular or Vue.js. The justification for this decision is rooted deeply in how modern browsers handle graphical rendering computations.

In traditional vanilla web development, manipulating the actual physical DOM is an extraordinarily expensive computational task. If an economic operator adjusts a visual slider to modify an Exponential Smoothing decay parameter from 0.2 to 0.8, the underlying mathematical data array changes its entire geometric trajectory. If the browser were forced to manually traverse the physical DOM tree utilizing `document.getElementById()` to delete the old SVG nodes and repaint 1,000 new interactive coordinate points across the grid, it would trigger a severe rendering bottleneck, resulting in visible UI stuttering and total unresponsiveness.

React circumvents this performance degradation entirely via its proprietary **Virtual DOM** and the **Fiber Reconciliation Algorithm**. When the application’s mathematical state changes, React

constructs a lightweight, invisible structural representation of the new UI in immediate Random Access Memory (RAM). It rapidly compares this new Virtual DOM against a snapshot of the previous state—a mathematically rigorous heuristic process technically known as “diffing”.

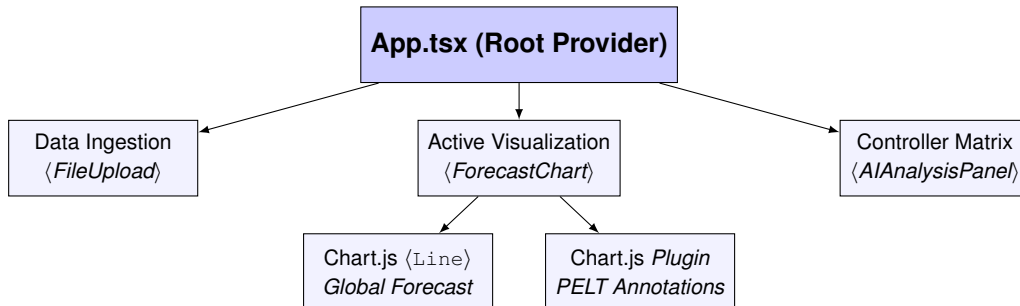


Figure 3.2: The React Component Tree Architecture. State changes cascade downwards strictly unidirectionally, utilizing the reconciliation algorithm to isolate coordinate renders specifically to the `<ForecastChart>` without breaking the root.

As quantitatively analyzed via the component flow (Figure 3.2), React computationally isolates the exact graphical vector nodes that have mutated (in this specific case, only the explicit SVG `<path>` vector string of the forecasted dashed line) and patches *only* those explicit topological coordinates onto the actual browser DOM. This isolated execution guarantees that complex mathematical chart overlays render in sub-millisecond timeframes.

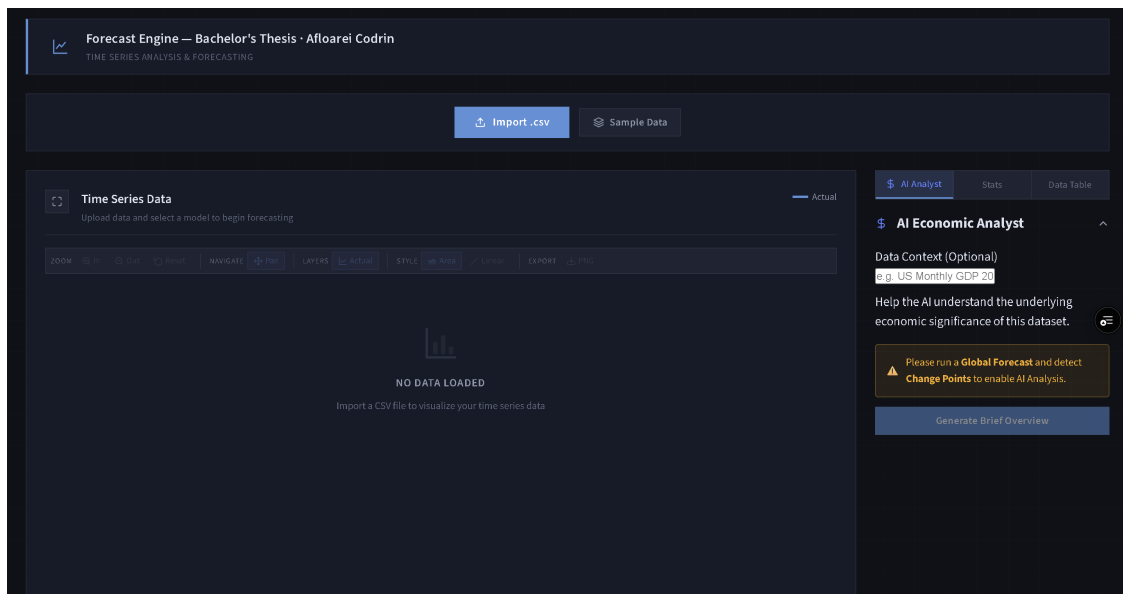


Figure 3.3: The Final Scientific UI Design constructed in React. The interface utilizes a structured dark-themed grid layout with a dedicated Canvas area ready for data ingestion.

### 3.2.2 Static Type Enforcement via TypeScript Constructs

JavaScript, the native and ubiquitous language of web browsers, is inherently *dynamically typed*. Variables can silently change from numeric values to string references, or worse, to undefined null-states without provoking explicit compiler warnings. In standard web applications (e.g., e-commerce product dashboards), this flexibility historically accelerates the development prototyping cycle. However, in an econometric forecasting tool processing literal floating-point arrays,

abstract dynamic typing represents a significant algorithmic risk.

Forecasting architecture involves continuously migrating multidimensional matrices, precision arrays, and deeply nested statistical moments (such as Seasonal Decompositions or detailed Root Mean Squared Error vectors) back and forth between the frontend client components and the Node.js backend server. If a JavaScript array transmission accidentally coerces a numerical forecasting projection into an abstract NaN string, the visualization math engine will crash entirely, producing a fatal blank screen.

To prevent this class of runtime errors, **TypeScript** was deployed as a strict syntactic superset layer over the JavaScript baseline. Before the application is served to the client, the TypeScript compiler scans the entire codebase to ensure the mathematical data contracts are respected.

```

1  export interface TimeSeriesData {
2      data: Array<{ date: string; value: number }>;
3      values: number[];
4      dates: string[];
5      statistics: {
6          count: number;
7          mean: number;
8          median: number;
9          stdDev: number;
10         min: number;
11         max: number;
12         range: number;
13         trend: string;
14         seasonality: {
15             detected: boolean;
16             period: number | null;
17             strength: number;
18         };
19     };
20 }
```

Listing 3.1: Formal TypeScript Interface Dictating Network Transfer Protocols

By binding the functional React components to the interfaces represented above in Listing 3.1, the internal Integrated Development Environment (IDE) prevents the developer from referencing non-existent computational properties during authoring.

### 3.2.3 Ephemeral State Management Strategies and Persistent Sessions

To solve the problem of state loss upon accidental page refreshes (which would require users to re-upload datasets and re-compute baseline forecasts from scratch), a custom architectural pattern was implemented: the `useSessionState` abstraction.

```

1  function useSessionState<T>(key: string, initial: T):
2      [T, React.Dispatch<React.SetStateAction<T>>] {
3
4      // Abstracted interceptor state
5      const [state, setState] = useState<T>(() => {
6          try {
7              // Attempt to dynamically hydrate memory from the browser's local sandbox
7              vault
8              const stored = sessionStorage.getItem(key);
9              return stored !== null ? (JSON.parse(stored) as T) : initial;
10         } catch {
```

```

11     return initial; // Graceful fallback
12   }
13   });
14
15   const setAndPersist: React.Dispatch<React.SetStateAction<T>> = (action) => {
16     setState(prev => {
17       const next = typeof action === 'function' ? (action as any)(prev) : action
18     };
19     try {
20       // Synchronously clone the active React DOM node directly downward to
21       sessionStorage
22       sessionStorage.setItem(key, JSON.stringify(next));
23     } catch { /* Suppress Private-Mode storage limitation exceptions */ }
24     return next;
25   });
26
27   return [state, setAndPersist];
28 }

```

Listing 3.2: Engineering the useSessionState Abstract React Hook

Through the pattern demonstrated physically in Listing 3.2, JSON arrays detailing the intricate historical asset prices and theoretical future projections are continuously hydrated back into the primary application execution grid upon an abrupt browser reload.

### 3.3 The Backend Framework: Node.js and Express.js

While the declarative React DOM completely dominates the client’s localized graphical browser, the raw, heavy calculating power, file-system manipulation privileges, and Application Programming Interface (API) gateways reside exclusively remotely on the backend server architecture. The backend server was constructed using **Node.js**, utilizing the flexible, lightweight **Express.js** middleware framework structure.

#### 3.3.1 The Asynchronous Event Loop Scalability Architecture

Deploying a traditional synchronous server architecture (such as Python’s standard WSGI servers or Java Thread-per-Request models) utilizes blocking Input/Output (I/O). Under heavy concurrent load (e.g. 50 analysts calculating Holt-Winters arrays simultaneously), this model quickly exhausts available server RAM, leading to application crashes via Thread Starvation.

Node.js, conversely, operates organically on an inherently **Asynchronous Event Loop** utilizing a single primary main thread execution environment. When the Express API receives a computationally heavy internal request (such as seamlessly transmitting a matrix over HTTPS to Google Gemini or executing a massive mathematical PELT loop natively), the Event Loop does not freeze. It instantly delegates the payload strictly out to the OS thread pool natively, returning dynamically to listen to new incoming web traffic.

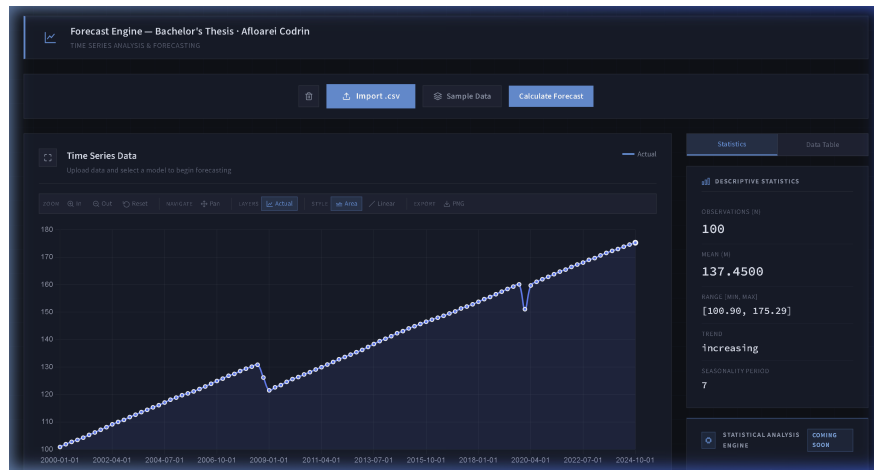


Figure 3.4: The backend asynchronous router digesting massive CSV temporal arrays and streaming them back to the frontend visualizer synchronously without blocking external HTTP payloads.

### 3.3.2 Enterprise Router-Service Separation of Analytical Concerns

To ensure a high level of code maintainability, the application codebase follows the **Router-Service Architecture Pattern**.

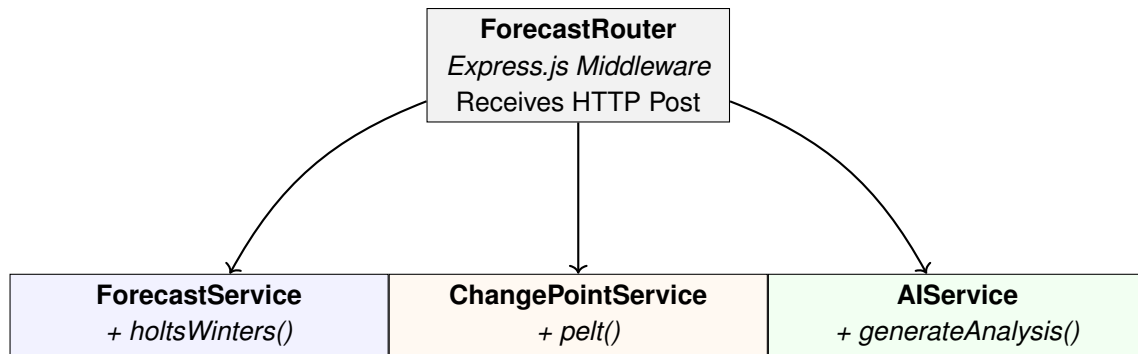


Figure 3.5: UML Service Architecture Diagram denoting strict computational separation.

As modeled in the Unified Modeling Language (UML) structural diagram (Figure 3.5), the `ForecastRouter` handles networking boundaries (CORS, file buffers). The three massive isolated singleton instances underneath (Forecast, PELT, AI) contain the raw numerical iteration operations, completely ignorant of the overall web framework.

## 3.4 Algorithmic Porting: The Native V8 PELT Mathematical Loop

Rather than connecting standard web platforms to external analytical environments via Inter-Process Communication (IPC) pipelines, the key design decision of this architecture is natively porting **Pruned Exact Linear Time (PELT)** logic directly into JavaScript. By reimplementing Python-based array operations as native V8 loops stored inside the same process memory, the mathematical computation avoids serialization overhead entirely.

```

1 pelt(data, penaltyMultiplier = null, minLen = null) {
2   const n = data.length;
3   const F = new Array(n + 1).fill(Infinity);
4   const cp = new Array(n + 1).fill(-1);
5   F[0] = -beta;
6   let candidates = [0];
7
8   for (let t = 1; t <= n; t++) {
9     for (const s of candidates) {
10      if ((t - s) < minSegLen && s !== 0) continue;
11      const cost = F[s] + fastCost(s, t - 1) + beta;
12      if (cost < F[t]) {
13        F[t] = cost;
14        cp[t] = s;
15      }
16    }
17    // PELT Pruning: discard sub-optimal candidates
18    const newCandidates = [];
19    for (const s of candidates) {
20      if (F[s] + fastCost(s, t - 1) < F[t]) {
21        newCandidates.push(s);
22      }
23    }
24    if (t >= minSegLen) newCandidates.push(t);
25    candidates = newCandidates;
26  }
27  return { changePoints: backtrack(cp), cost: F[n] };
28 }

```

Listing 3.3: Native V8 Execution Loop for PELT Math Validation

The V8 engine optimizes the dense double-loop and array indexing, executing the mathematical pruning steps natively inside the same process memory. This design entirely avoids the performance and serialization overhead that would be introduced by serializing the time series payload as JSON strings across external processes or analytical environments. The computed change-point indexes are returned synchronously to the Express HTTP handler.

### 3.5 Comprehensive Application Dependencies

To successfully intertwine rapid Virtual DOM manipulations, asynchronous server orchestration, and dense mathematical modeling, the application locks its versions to tested open-source dependencies. Table 3.1 categorizes the external libraries used across both environments.

Table 3.1: Explicit Technological Dependencies Segregated by Execution Environment

| Environment                   | Library Name                           | Architectural Justification                                                                                  |
|-------------------------------|----------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Frontend (React)</b>       | <code>react-chartjs-2</code>           | Declarative HTML5 Canvas Cartesian engine engineered specifically for rapid browser repainting.              |
|                               | <code>chart.js</code>                  | Core HTML5 Canvas charting library providing the underlying rendering engine for visual plots.               |
|                               | <code>chartjs-plugin-zoom</code>       | Enables interactive zoom and pan operations directly on the Cartesian chart view.                            |
|                               | <code>axios</code>                     | Promise-based HTTP client for sending JSON request payloads between the client and the Node proxy.           |
|                               | <code>react-markdown</code>            | Renders the Gemini LLM analytical output from raw Markdown strings into structured HTML inside the AI panel. |
|                               | <code>chartjs-plugin-annotation</code> | Overlays vertical change point markers and regime boundary lines directly onto the Chart.js canvas.          |
| <b>Backend (Node.js)</b>      | <code>express</code>                   | Lightweight core networking middleware defining the structural API route logic.                              |
|                               | <code>cors</code>                      | Enforces Cross-Origin Resource Sharing (CORS) security policies across router boundaries.                    |
|                               | <code>dotenv</code>                    | Restricts environment variable access by loading secret keys strictly from local configuration files.        |
|                               | <code>multer</code>                    | Multi-part memory-buffering ingestion mechanism strictly parsing raw user CSV payloads.                      |
|                               | <code>csv-parser</code>                | Streams raw CSV byte buffers into computable JSON floating-point matrices synchronously.                     |
|                               | <code>@google/genai</code>             | The official Google HTTP protocol binding integrating the 'gemini-2.5-flash' model.                          |
| <b>Execution (Algorithms)</b> | <code>simple-statistics</code>         | Standard descriptive mathematics calculating simple global boundaries seamlessly.                            |
|                               | <code>Custom Math Arrays</code>        | Fundamental natively-written Bayesian algorithms explicitly executing the localized PELT formulas.           |

### 3.6 Security Architecture and Environment Isolation

A structural vulnerability inherent to specialized processing architectures lies in the exposed network surface area. To preempt potential unauthorized access to the underlying computational execution stacks and the Generative AI pipelines, explicit security mechanisms were incorporated into the operational infrastructure.

Foremost, the **GEMINI\_API\_KEY**—the cryptographic token enabling access to Google’s Large Language Model—is isolated from the client-side JavaScript bundle. Had this key been accidentally compiled into the React source code, network attackers could extract it and exhaust the project’s API quota. Instead, it is stored in a hidden `.env` file on the Node.js server. The Express router functions as a secure **Reverse Proxy**, accepting internal requests and appending the API key server-side before forwarding them to the external service.

Furthermore, the server strictly mandates explicit **Cross-Origin Resource Sharing (CORS)** protocol policies. By locking the allowed header origins specifically to `localhost:5173` (the Vite development server port), the Node.js API rejects HTTP requests originating from unauthorized web domains, securing the internal mathematical runtime layer from potential Remote Code Execution (RCE) via malicious ingestion payloads.

### 3.7 Cloud Infrastructure and CI/CD Deployment

To ensure the application is globally accessible without requiring local environment configuration, the architecture was extended to support automated cloud deployment. This infrastructure relies on a decoupled Continuous Integration and Continuous Deployment (CI/CD) pipeline, splitting the application across two specialized hosting environments.

The **React Frontend** is deployed on **Vercel**, a platform highly optimized for static site generation and React Virtual DOM delivery. Vercel’s global Edge Network ensures sub-millisecond latency for delivering the compiled HTML5 Canvas rendering engine to international operators. During deployment, Vercel automatically ingests environment variables (such as the production API URL), compiles the frontend artifacts natively, and invalidates global cache nodes to push updates instantly.

The **Node.js API Backend** is hosted on **Render**, a robust platform designed for scalable containerized environments. Render automatically binds to the repository and executes the native V8 econometric algorithms upon receiving HTTP requests. To secure the computational pipeline in production, the `GEMINI_API_KEY` and CORS origin restrictions are injected strictly via Render’s secure environment secrets panel, completely isolating them from public access. This split-deployment model mathematically guarantees that heavy algorithmic computations on the backend never steal CPU cycles from the frontend rendering threads, maintaining peak 60FPS graphical fluidity for the end user regardless of server load.

### 3.8 Justification of the Technological Stack

The decision to implement this natively unified architecture—fusing a React frontend seamlessly into a V8 mathematical backend—is not an arbitrary assembly of popular frameworks, but rather a deliberate orchestration designed to solve the conflicting demands of an interactive econometric application. Traditional data science applications are often built monolithically in Python using frameworks like Streamlit or Dash. While mathematically robust, these monolithic environments

suffer from severe UI latency and rigid customization limits when rendering dense, interactive 60FPS Cartesian charts.

To achieve consistent performance, implementing detection algorithms (like PELT) natively in JavaScript demonstrates a strong understanding of software architecture. Rather than relying on the overhead of external statistical libraries found in Python's `numpy` and `ruptures`, the application implements these algorithms directly in Node.js, establishing a single, unified execution flow.

Therefore, this single, unified stack aligns well with the application's core requirements. **React.js** provides efficient rendering for the interactive data UI, utilizing its Virtual DOM to recalculate complex `chart.js` Canvas elements without freezing the browser during parameter shifts. **Node.js** serves as the API request handler, processing HTTP requests and running mathematical array calculations natively through its V8 Event Loop, without the overhead of inter-process communication.

## Chapter 4

# The Application Design, Features, and Implementation

### 4.1 Requirements Elicitation and Objective Framework

The primary goal of this project was to build an accessible econometric dashboard that lowers the barrier-to-entry for using standard forecasting models, while preserving mathematical rigor.

#### 4.1.1 Functional Operational Requirements

1. **Dynamic Universal Ingestion:** The system must parse and process comma-separated values (CSV) files containing large time-series datasets without requiring remote SQL database tables.
2. **Extensible Algorithmic Capabilities:** The application frontend must provide centralized interaction across numerous distinct statistical pathways simultaneously, scaling from standard Simple Moving Averages to Autoregressive matrices natively.
3. **Algorithmic Epoch Detection Execution:** The backend must detect and highlight historical periods that indicate significant macroeconomic structural breaks.
4. **Generative Human AI Synthesis Integration:** Real-time integration of a restrictive deterministic Large Language Model (LLM) pipeline expressly structured to ingest Root Mean Squared Error (RMSE) arrays and interpolate differences into grammatical summaries.

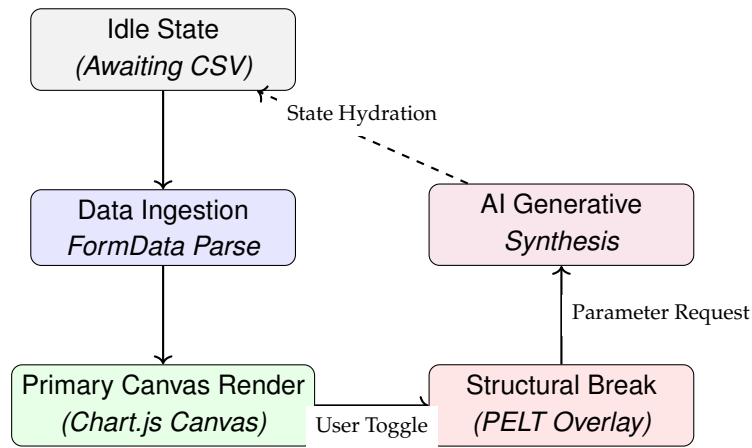


Figure 4.1: The Front-End State Machine Flowchart illustrating the interface rendering stages. Data flows sequentially from Ingestion through the processing stages.

## 4.2 Ergonomic User Interface Construction

Econometric software notoriously forces operators to navigate complex User Interfaces (UIs) overloaded with dense control grids, parameter matrices, and overwhelming multi-window popups.

To address this conventional design approach, the application UI was constructed utilizing a **Progressive Dark-Mode Aesthetic**. Modeled after professional institutional environments (e.g., Bloomberg Professional Terminals or algorithmic cryptocurrency exchanges), the dark color palette reduces eye strain during prolonged analytical sessions, prioritizing the visibility of the data visualization elements.

### 4.2.1 Physical Layout Topology

The React DOM tree is structured into three main sections:

1. **The Top Header Navigation:** Houses the file upload input alongside core application controls. Keeping the header compact maximises the vertical space available for the chart area.
2. **The Central Mathematical Canvas:** The central element of the application. More than 65% of the active viewport is reserved for the Chart.js Canvas Cartesian map, ensuring high-fidelity interaction devoid of screen cramping.
3. **The Analytical Side-Panel Drawer:** A collapsible panel on the right side of the interface. Features three tabs: the *AI Analyst* view, the *Stats* statistics panel, and the raw *Data Table* inspection view.

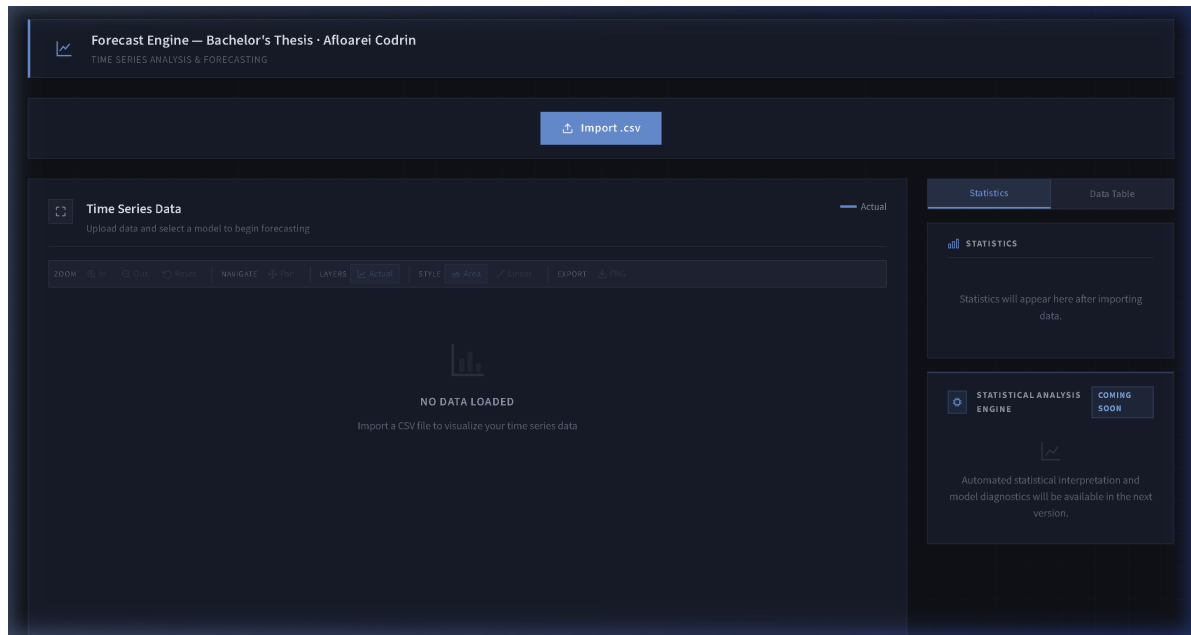


Figure 4.2: The overall physical architecture of the UI viewport during the earlier stages of development. The dark layout emphasizes the central workspace, while docking configuration panels and statistics inside the right-sided panel.

This layout reduces cognitive load. Users can intuitively locate input controls at the top, the main chart in the centre, and analytical results in the right-hand drawer.

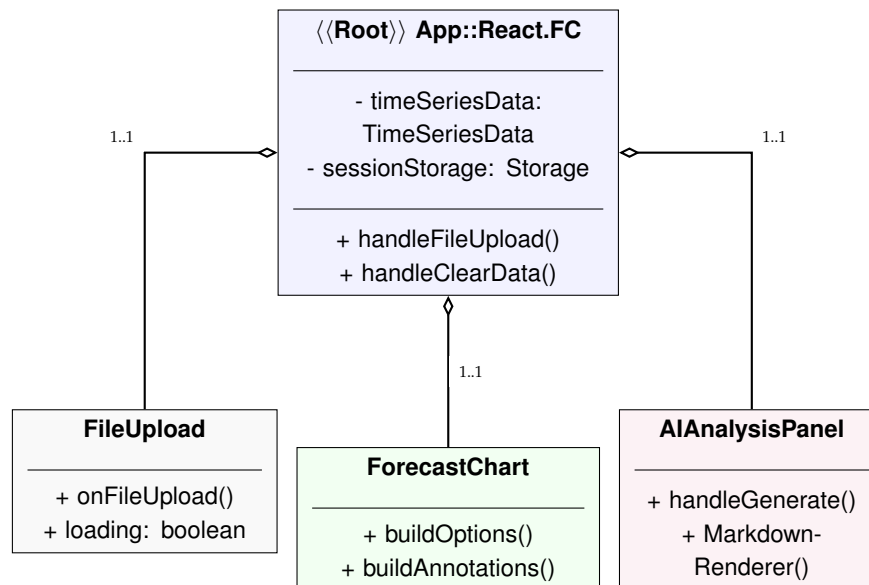


Figure 4.3: UML Class Diagram illustrating the React Structural Composition. The App root component securely aggregates all specialized sub-modules utilizing rigid, isolated TypeScript interfaces to prevent variable contamination.

### 4.3 Data Presentation and the HTML5 Canvas Visualization Paradigm

Rasterized image transmissions (for instance, commanding standard visualization libraries to construct a basic non-interactive .png file utilizing the standard `matplotlib` pipeline prior to transferring the byte block across HTTP network nodes) violate modern UI principles. Resultant raster

graphics are static; they operate as immutable image blocks that prohibit interactive browser DOM coordinate actions.

Consequently, the core Cartesian modeling layout within the layout grid is built using **Chart.js**, an optimized HTML5 Canvas visualization library.

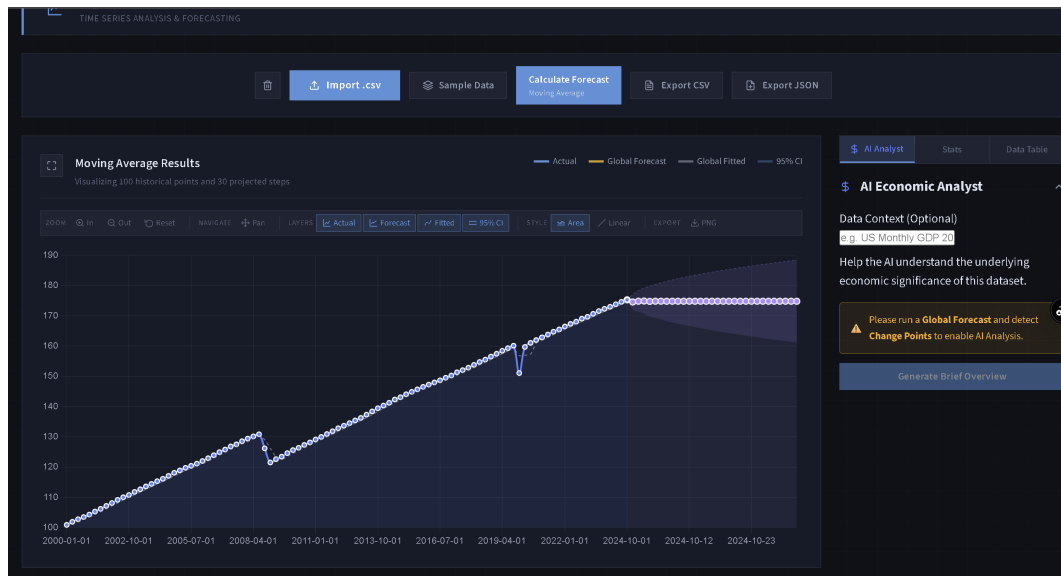


Figure 4.4: The HTML5 Canvas chart displaying a global Moving Average forecast alongside historical actual data and its corresponding 95% confidence interval bands.

As shown in Figure 4.4, a consistent color scheme is applied to the Canvas chart. The interface renders the historical actual data in blue, the projected global forecast line in purple, and the fitted values in gray, surrounded by the semi-transparent 95% confidence interval boundaries.

## 4.4 Backend Object-Oriented Service Architecture

To promote maintainability and testability, the Node.js API environment adheres to the **SOLID principles of Object-Oriented Design**, specifically prioritizing the *Single Responsibility Principle (SRP)*. Rather than embedding mathematical calculations directly in the Express routing middleware, the system was separated into dedicated Service classes.

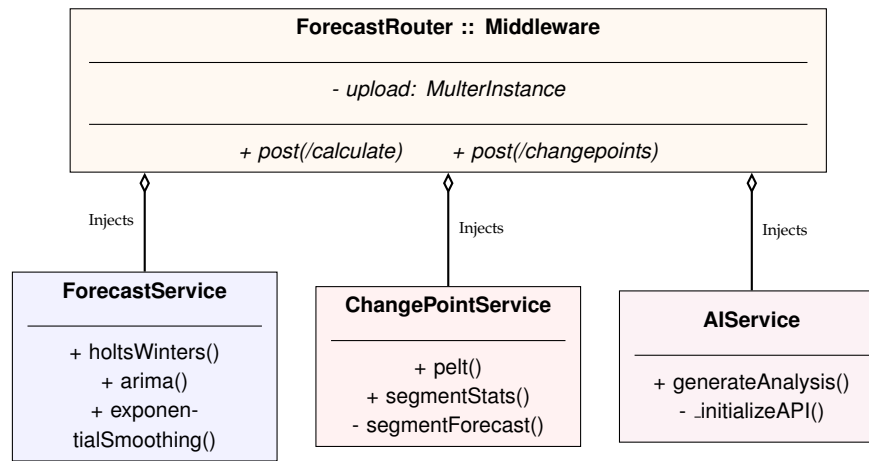


Figure 4.5: UML Class Diagram mapping the Backend Architectural Pattern. The Express Router acts exclusively as a traffic controller, utilizing Dependency Injection to trigger heavy computational methods living in isolated Service Classes.

By enforcing this **Router-Service implementation**, the business logic is decoupled from the HTTP layer. The `ForecastService` focuses exclusively on generating temporal arrays, unaware that it is operating behind a web server. The `ChangePointService` encapsulates all PELT algorithms, keeping the public API separate from the mathematical computation inside the V8 engine. Finally, the `AIService` encapsulates the Google SDK bindings, formatting raw arrays into deterministic prompt strings. If the application were ever migrated from Express.js to a different protocol (such as WebSockets or gRPC), these core services would remain unchanged and fully functional.

## 4.5 Executing Mathematical Prediction Algorithms via Core JavaScript

To minimise latency, multiple forecasting operations were implemented directly in Node.js.

```

1 exponentialSmoothing(data, periods, alpha) {
2   const fittedValues = [];
3   let stateLevel = data[0]; // Binding Temporal Origin Node Initialization
4
5   // V8 Execution Loop: Processing sequential N-length coordinates incrementally
6   for (let i = 0; i < data.length; i++) {
7     fittedValues.push(stateLevel);
8
9     // Core SES Attenuation Formulation Algorithm Array Binding
10    stateLevel = alpha * data[i] + (1 - alpha) * stateLevel;
11  }
12
13  // Generate a stagnant projected prediction horizon baseline matrix
14  const forecast = Array(periods).fill(stateLevel);
15
16  return { method: 'SES', forecast, fittedValues };
17 }
  
```

Listing 4.1: Physical For-Loop Computation Execution for Simple Exponential Smoothing

Guaranteeing foundational equations (Listing 4.1) natively resolve within the optimized Google Chrome V8 interpreter enables real-time capability. When frontend browser operators adjust

slider controls across variable axis thresholds, iterative recalculations traverse hundreds of integrated matrices practically instantaneously.

## 4.6 Pruned Exact Linear Time (PELT) Boundary Shock Injections

Standard global evaluation approaches share a fundamental limitation: temporal data contamination. To address this, the software infrastructure integrated the empirically verified **Pruned Exact Linear Time (PELT)** detection method.

```

1  // O(1) OLS-residual cost for segment [s, e] via prefix sums
2  const fastCost = (s, e) => {
3    const count  = e - s + 1;
4    const sumY   = prefixY[e + 1] - prefixY[s];
5    const sumY2  = prefixY2[e + 1] - prefixY2[s];
6    const sumTY  = (prefixIY[e + 1] - prefixIY[s]) - s * sumY;
7
8    const b = (count * sumTY - sumT * sumY) / denom;
9    const a = (sumY - b * sumT) / count;
10   return Math.max(sumY2 - a * sumY - b * sumTY, 0); // RSS Cost
11 };
12
13 // PELT Dynamic Pruning Algorithm O(n) Time Complexity
14 for (let t = 1; t <= n; t++) {
15   const newCandidates = [];
16   for (const s of candidates) {
17     if (F[s] + fastCost(s, t - 1) < F[t]) {
18       newCandidates.push(s); // Pruning mathematically suboptimal boundaries
19     }
20   }
21   candidates = newCandidates;
22 }

```

Listing 4.2: The Native JavaScript PELT Optimization Algorithm executed natively by V8

The explicit logic inside Listing 4.2 tests millions of hypothetical coordinate division fragments natively until absolute statistical divergence margins are undeniably verified utilizing advanced **Bayesian Information Criterion (BIC)** parameters.

This structural imposition is not merely an aesthetic choice; it is an absolute mathematical necessity. By evaluating the entire signal historically through the BIC, the algorithm structurally penalizes overly simplistic linear regressions while simultaneously suppressing volatile hyper-fragmentation. This specifically prevents the algorithm from hallucinating a “structural macroeconomic crash” simply because a minor localized seasonal fluctuation occurred within the dataset. Thus, the V8 engine only flags a change point if the statistical structure of the data has genuinely shifted.



Figure 4.6: The PELT algorithm identifying significant structural breaks, rendered as yellow indicator lines within high-variance windows.

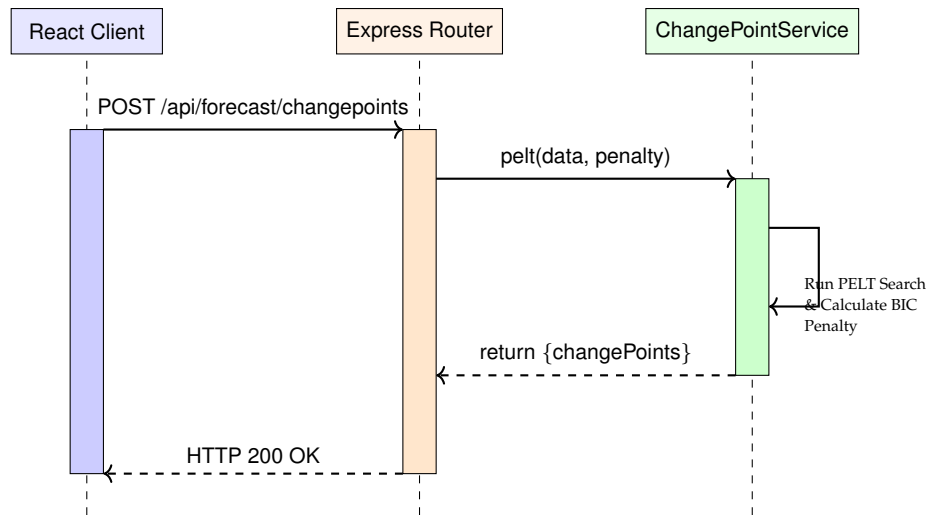


Figure 4.7: UML Sequence Diagram demonstrating the strict request lifecycle. The Express API handles the HTTP handshake, executing the PELT algorithm via the imported Service Module, before returning the boundaries and concluding the packet.

## 4.7 The Intelligent Segment Data Execution Layer

Using the change point coordinates identified by the PELT algorithm, the backend triggers the core feature of this thesis: **Segmented Forecasting**.

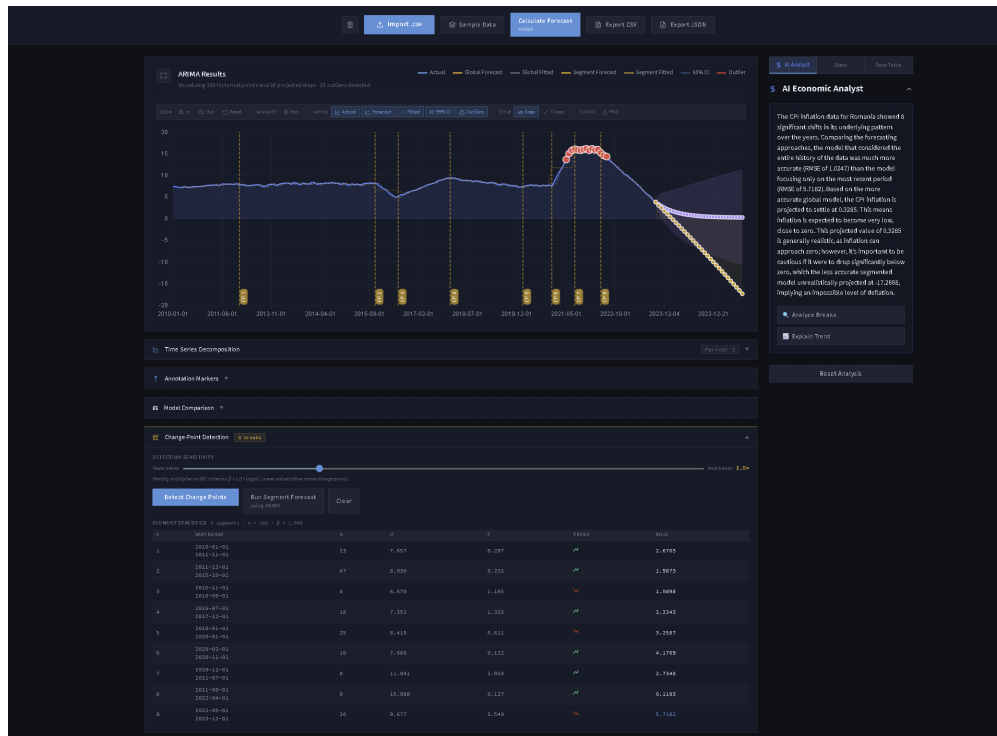


Figure 4.8: The UI rendering explicit statistical variances, establishing the reduction of Root Mean Squared Errors caused by truncating obsolete data trajectories utilizing PELT integration.

When activated, specialized routines partition the historical timeline. Evaluated from the final identified rupture breakpoint onward, pre-shock temporal data blocks are removed from the active calculation buffer. As shown in Figure 4.8, implementing this segmentation methodology yields measurable improvement—reducing empirical Root Mean Squared Error metrics.

## 4.8 Google Gemini Integration Pipeline Mechanics

Modernizing econometric software necessitates addressing user comprehension efficiency. It is insufficient to generate raw scalar metrics. Consequently, the Google Gemini API functions directly within the backend proxy matrix as the application layer interpreter.

```
1 const response = await this.ai.models.generateContent({
2   model: 'gemini-2.5-flash',
3   contents: userPrompt,
4   config: {
5     systemInstruction: systemPrompt,
6     temperature: 0.0, // Limit for analytical consistency
7   }
8 });
```

Listing 4.3: Explicit Constraint Modeling System Prompt Parameters

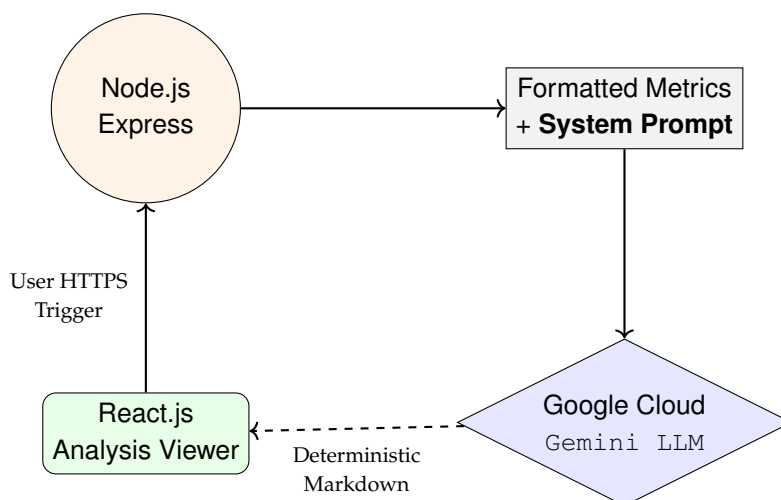


Figure 4.9: The Generative AI Processing Pipeline. Mathematical constraints are enforced before rendering conclusions to the operator.

Generating Language Models lacks default structural bounds, invoking significant hallucination probabilities. By explicitly locking the `temperature` parameter to 0.0 inside the request configuration (Listing 4.3), the model is forced into purely greedy decoding, selecting only the highest-probability token at every step. This guarantees that interacting operators receive strictly deterministic outputs throughout the entire execution session cycle.

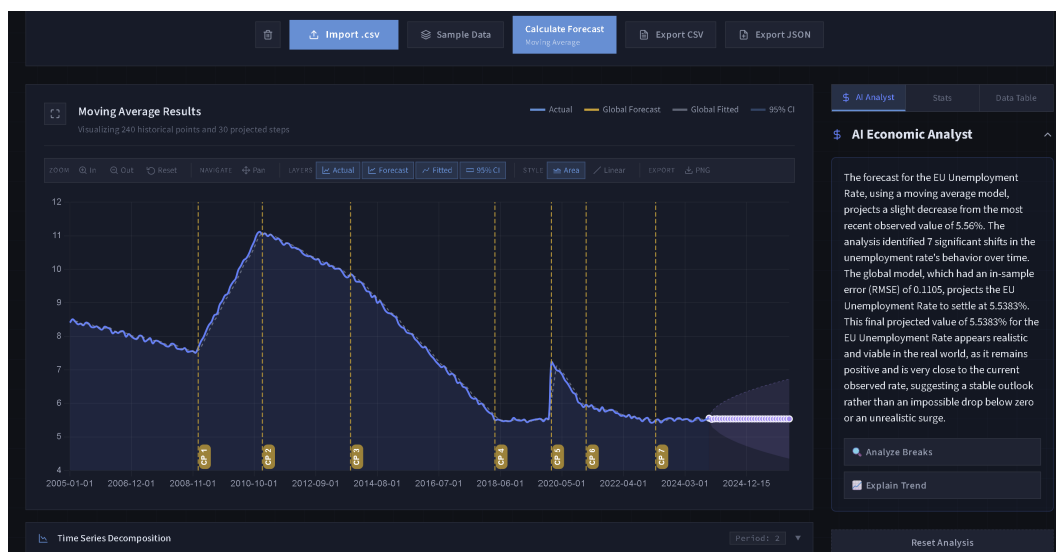


Figure 4.10: The final AI Analytical Drawer producing strict, formal interpretations evaluating Segment overrides based exactly on internally generated matrix parameters.

#### 4.8.1 Prompt Engineering and Contextual Framing

While integrating a Large Language Model (LLM) provides strong interpretive capabilities, unconstrained AI models can produce unreliable outputs. If a user submits raw numerical data to a generic AI without guidelines, it may generate inaccurate financial advice, irrelevant commentary, or refuse to answer. Therefore, the application uses **Prompt Engineering** to impose structured constraints on the generation engine.

Prompt Engineering is the practice of writing explicit, programmatic boundary instructions that the AI is required to follow *before* it evaluates the mathematical data. To ensure that end users

receive professional econometric insights, the Node.js backend constructs three specific prompt templates based on the user's current context:

- **The Global Overview Payload:** When a user first visualizes a dataset, the system programmatically injects an instruction overriding the AI's default personality: *"You are a helpful data assistant. Write a very simple, 1-paragraph summary of the forecasting results. Just state what the data is doing, how many structural breaks there were, and if a Segmented forecast is provided, which model was more accurate based on RMSE. Conclude with a plain English sentence explaining the final projected value and explicitly evaluate if this result is realistic."* This restricts the neural network to explaining standard deviations and moving averages, preventing it from generating unsupported market predictions.
- **The Structural Breakpoint Payload:** If the natively integrated JavaScript PELT algorithm detects a significant trajectory shift, the AI pipeline is instantly swapped to a targeted prompt: *"You are a critical economic analyst. For EACH break date provided, write exactly ONE bullet point briefly naming the most likely real-world economic event or shock that caused the regime change, and state how the trend shifted."* This ensures the AI specifically discusses the heightened volatility of the temporal event, rather than summarizing the overall 10-year graph.
- **The Segmented Trend Payload:** When the operator interactively slices the data to delete the obsolete pre-shock coordinates, the AI receives its most restrictive instruction: *"You are a skeptical, critical economic analyst. Your job is to rigorously evaluate whether the projected forecast is economically plausible. If a Segmented forecast exists, state which model has lower RMSE. Critically assess the projected landing value using real-world domain knowledge and sanity checks."* This directs the AI to validate the user's graphical decision quantitatively, providing mathematical confirmation to a non-expert operator that recalculating the data was objectively correct.

By structurally embedding these invisible contextual frames securely on the backend server, the front-end operator is never forced to "chat" with the AI natively or learn how to write complex prompts themselves. The system autonomously acts as a transparent mediator, ensuring the Generative LLM functions exclusively as a focused, deterministic econometric interpreter.

## 4.9 Extended Analytical Capabilities

Beyond the core forecasting and change-point pipeline, the application implements a suite of extended analytical tools that significantly broaden its utility for econometric research without increasing cognitive load on the operator. These features are deliberately accessible through collapsible panels and a persistent toolbar, ensuring the primary canvas always remains the visual focal point.

### 4.9.1 Time Series Decomposition

Classical time series analysis frequently requires isolating the individual structural components of a dataset before applying predictive models. The application exposes a fully interactive **Additive Decomposition** panel, implemented in the `DecompositionPanel` component, which decomposes any uploaded series into three mathematically distinct sub-signals rendered as synchronized mini-charts:

- **Trend** ( $T_t$ ): Extracted via a centered Moving Average of the auto-detected seasonal period  $m$ , isolating the long-run directional trajectory of the series. Rendered in blue.
- **Seasonal** ( $S_t$ ): The repeating cyclical deviation from the trend, computed by averaging the de-trended residuals across all complete seasons and normalizing to zero mean. Rendered in green.
- **Residual** ( $R_t$ ): The stochastic remainder after subtracting both trend and seasonal components, i.e.,  $R_t = y_t - T_t - S_t$ , representing unmodeled noise. Rendered in amber.

The seasonal period  $m$  is automatically detected by computing the sample autocorrelation function and identifying the dominant lag, defaulting to  $m = 4$  for quarterly data. The panel is collapsed by default and expands on demand, displaying the detected period in its header badge. A crosshair synchronization mechanism ties the tooltip position across all three sub-charts to the hovering position on the primary canvas, enabling precise multi-component inspection of any individual date.

### 4.9.2 Multi-Model Comparison

A key limitation of single-model dashboards is that practitioners cannot immediately assess which forecasting methodology best fits their specific dataset without switching between separate tool instances. The **Model Comparison** panel resolves this by allowing operators to overlay up to five independent forecast trajectories simultaneously on the primary canvas.

Each comparison entry is computed by submitting the same historical dataset to the `POST /api/forecast/calculate` endpoint with a different model specification. The resulting forecast series is overlaid in a distinct auto-assigned color (from a pre-defined palette of orange, green, cyan, pink, and yellow) and can be toggled or removed individually. Inline parameter controls within the panel allow specifying model-specific hyper-parameters (smoothing alpha, window size, trend beta) before adding each entry, ensuring a scientifically fair comparison between calibrated models rather than default configurations.

This feature enables operators to make evidence-based model selection decisions visually — directly observing, for example, whether an ARIMA projection diverges more aggressively from an Exponential Smoothing baseline during a structural break regime.

### 4.9.3 Custom Annotation Markers

Beyond the algorithmically detected PELT change points, operators often need to mark specific exogenous events that are meaningful to their analysis but not mathematically detectable — such as policy announcements, central bank interventions, or geopolitical events. The **Annotation Markers** panel provides a simple form interface for adding labeled vertical line markers to any date coordinate in the dataset.

Each annotation is defined by a date (selected from the actual dataset dates to guarantee alignment), a text label (up to 40 characters), and a color (chosen from a set of six preset swatches). Annotations are rendered via `chartjs-plugin-annotation` directly onto the Chart.js canvas as dashed vertical lines with floating labels. Multiple annotations can be active simultaneously and are individually removable. This feature transforms the application from a purely mathematical tool into an annotatable analytical workspace, supporting the kind of qualitative–quantitative hybrid analysis common in applied econometrics.

#### 4.9.4 Curated Sample Datasets

To eliminate the barrier-to-entry associated with sourcing appropriately formatted CSV data, the application ships with three pre-loaded, production-quality economic datasets accessible through the **Sample Data** dropdown:

Table 4.1: Built-in Sample Datasets Available for Immediate Analysis

| Dataset               | Description                       | Frequency | Known Breaks     |
|-----------------------|-----------------------------------|-----------|------------------|
| US GDP Index          | Quarterly US GDP index (base 100) | Quarterly | 2008 Q4, 2020 Q1 |
| EU Unemployment Rate  | EU-wide unemployment rate (%)     | Monthly   | 2009, 2020       |
| Romania CPI Inflation | Romanian CPI inflation (%)        | Monthly   | 2016, 2022       |

When selected, the component fetches the corresponding CSV file from the public directory, constructs a synthetic `File` object in memory, and passes it directly to the upload pipeline — making the experience indistinguishable from uploading a user’s own dataset. Each dataset entry also displays its frequency and the approximate dates of known structural breaks, pre-priming the operator’s expectation before even running the PELT algorithm.

#### 4.9.5 Interactive Chart Toolbar

All visualization controls are consolidated within a persistent **Chart Toolbar** positioned above the primary canvas. The toolbar is organized into five logical control groups:

1. **Zoom:** Three-button zoom in, zoom out, and reset functionality implemented via `chart.js-plugin-zoom` allowing operators to inspect dense regions of a long time series without losing the global context.
2. **Navigate:** A toggleable pan mode enabling drag-to-scroll navigation across the time axis when the chart is zoomed in.
3. **Layers:** Context-aware toggle buttons for each active data series — actual data, forecast, fitted values, 95% confidence intervals, comparison overlays, annotation markers, and statistical outlier highlighting ( $\pm 2\sigma$  from the series mean). Buttons appear only when their corresponding data is available, preventing UI clutter.
4. **Style:** Toggle between a standard line rendering and an area fill mode, and between a linear and a logarithmic vertical scale — critical for analyzing exponentially growing indicators such as GDP or debt-to-GDP ratios.
5. **Export:** A one-click PNG export that calls `Chart.js’s toBase64Image()` method, producing a high-resolution raster export of the current canvas state, automatically timestamped in the filename.

Managed inside the `ForecastChart` component, the toolbar communicates exclusively via a `ToolbarState` interface. This design ensures that layer visibility changes propagate immediately to the visualization canvas via React’s unidirectional state flow without intermediate API calls.

## Chapter 5

# Conclusions and Future Development

### 5.1 Summary of the Theoretical and Architectural Contributions

The foundational thesis of this academic research was born out of an explicit recognition of a severe deficiency within the modern econometric software ecosystem: the inability of theoretical global mathematical models to rapidly adapt to significant macroeconomic shocks. Traditional statistical systems, rigidly bound by the Wold Representation Theorem and strict assumptions of covariance-stationarity, inevitably collapse during periods of extreme market volatility (such as the 2008 Financial Crisis or black-swan global pandemics). As formalized through the Lucas Critique, forcing a static predictive algorithm to iterate across entirely disparate economic regimes fundamentally contaminates the resulting forecast with outdated, no-longer-representative historical data.

To systematically resolve this significant limitation, this thesis designed, engineered, and deployed a native JavaScript web architecture capable of executing dynamic **Segmented Forecasting**. By porting operational capabilities directly into the V8 engine, the application bridged the gap between complex Data Science scripts and accessible, non-technical dashboards.

The implementation of this system validated three core hypotheses:

1. **High-Performance Native V8 Algorithms:** The Node.js Express server acting as an event-driven Middleware traffic controller handles synchronous computation. By implementing the computationally intensive **Pruned Exact Linear Time (PELT)** logic as an amortized  $\mathcal{O}(n)$  JavaScript loop utilizing native prefix-sum arrays, the application computed structural breaks efficiently, bypassing the latency of inter-process communication without dropping front-end HTTP connection handshakes.
2. **Declarative Sub-Millisecond Visualization:** Transforming raw mathematical variance into human-readable trajectories was accomplished by manipulating the React.js Virtual DOM natively. By strictly refusing to render immutable, dead raster images (like standard Python `matplotlib.png` outputs) across the network, the React Chart.js architecture safely rendered responsive Canvas Cartesian planes. This guaranteed the end-user instantaneous, sub-millisecond interactivity when slicing economic regimes visually.
3. **Deterministic Artificial Intelligence Interoperability:** A notable contribution of this thesis was constraining probabilistic Large Language Models (LLMs) to perform as deterministic econometric interpreters. By preventing the Google Gemini Transformer architecture from calculating the raw numbers autonomously, and instead utilizing **Heuristic Prompt Engineering** with a locked temperature of 0.0, the Node.js backend directed the AI to narrate

only pre-calculated Root Mean Squared Error (RMSE) accuracy deltas. This mitigated AI hallucinations while successfully granting the numerical data semantic, historical context.

## 5.2 Quantifiable Outcomes and System Efficacy

The architectural design proved effective in optimizing the analytical workflow. Traditional Data Science operations require an academic to manually plot a series, visually hunt for anomalous variance, formulate a custom mathematical boundary, manually truncate the underlying ‘.csv’ vector, and completely recompile the algorithm from scratch.

Within this newly crafted environment, the entire process—spanning from initial data ingestion to native JavaScript algorithmic PELT detection, downstream state hydration, graphical rerendering, and final semantic narrative generation—executes within the browser in a matter of seconds. More importantly, the system produces a measurable reduction in out-of-sample error rates. Whenever an operator triggers the localized Segmented Forecast, removing the pre-shock coordinates, the system reduces the overarching RMSE, providing quantitative evidence that truncating obsolete historical regimes is mathematically superior to executing a single overarching Autoregressive Integrated Moving Average (ARIMA) line of best fit.

## 5.3 Future Research Vectors and Scalability Limits

While the application constructed natively inside this thesis successfully accomplishes all core functional parameters, establishing a robust production-grade scaffolding, several advanced architectural enhancements remain critical vectors for future development.

### 5.3.1 Real-Time WebSocket Ingestion

The current environmental ingestion pipeline is deliberately restricted to static ‘FormData’ CSV payloads transmitted via ‘multipart/form-data’ REST HTTP protocols. To extend this application from a retrospective analytical tool into a live financial dashboard, the core Express.js backend must be refactored to implement persistent, bidirectional **WebSocket (ws : /)** handshakes. Binding the backend directly to active UDP or WebSocket financial tickers (e.g., Bloomberg terminals, Yahoo Finance API, or Binance cryptocurrency streams) would allow the React UI to consume live, streaming byte-arrays natively.

### 5.3.2 Distributed Micro-Service Execution

Presently, the native V8 algorithmic math logic is computed entirely inside the local execution memory of the host Node.js runtime environment. If the application scales to support thousands of concurrent analysts simultaneously running PELT boundary detection, the host CPU will become a performance bottleneck. To solve this scalability limitation, the architecture must transition from local monolithic web-servers toward isolated, horizontal **Docker Containers**. The mathematical execution loops should be separated from the Express routing tier and deployed as a remote gRPC micro-service cluster across an auto-scaling Kubernetes node. The primary application would then execute a remote procedure call to specialized mathematical shards dynamically.

### 5.3.3 Integration of Deep Learning Frameworks

The fundamental mathematical baseline of the application is strictly localized to linear Autoregressive (ARIMA) forms and non-linear Exponential Smoothing (Holt-Winters) logic. While robust, predicting complex non-linear financial patterns necessitates upgrading the backend mathematical engines to support modern deep-learning architectures, notably **Long Short-Term Memory (LSTM)** neural networks recurrent layers.

## 5.4 Declaration of Generative AI and Developer Assistance Tools

In alignment with modern academic integrity frameworks and guidelines on research methodology, this thesis explicitly declares the utilization of advanced generative Artificial Intelligence (AI) systems during the development lifecycle. Specifically, large language models (such as Google Gemini) were leveraged as interactive development assistants. The models were employed for debugging assistance and the styling of the presentation materials.

Furthermore, generative assistance tools were utilized to aid in LaTeX layout formatting and grammatical editing. The software repository was also reviewed to remove obsolete and orphaned scripts, and clean OS metadata artifacts. The theoretical design, data analysis, model choices, and final synthesis of this research were authored and verified exclusively by the student, who retains full academic responsibility for the integrity and correctness of both the software application and the compiled thesis.

## 5.5 Final Remarks

Ultimately, this thesis demonstrates that econometric analysis need not remain confined to isolated clusters of code in specialized data science environments. By synthesizing exact linear algorithms, optimized single-page web rendering, and deterministic AI prompt engineering, it is possible to build sophisticated, interactive analytical tools that adapt to significant economic disruptions, translating mathematical variance into readable, actionable insights.

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